

Q1. Bayes Rule Applied to Text (based on slide: Bayes' Rule for documents)

The PPT shows that classification is based on:

$$c_{MAP} = \arg \max_{c \in C} P(c) P(d | c)$$

Tasks:

1. Explain in your own words what each term means: $P(c)$, $P(d|c)$ and $P(c|d)$.

Ans:

$P(c)$ (the prior probability of class c):

This is how likely a class is *before seeing the document*.

Example: If we're classifying emails into *spam* vs *not spam*, and historically 20% of all emails are spam, then $P(\text{spam})=0.2$.

$P(d|c)$ (the likelihood):

This is how likely it is to see the document d *given* that it belongs to class c .

Example: If spam emails often contain words like "free" or "winner," then a document with those words has a higher $P(d|\text{spam})$.

$P(c|d)$ (the posterior probability):

This is what we really want: the probability that the document belongs to class c after observing its words.

Example: Given an email with "free lottery winner," $P(\text{spam}|d)$ will be very high.

2. Why can the denominator $P(d)$ be ignored when comparing classes?

Bayes' Rule says: $P(c|d)=P(d)P(d|c)/P(d)$

$P(d)$ = the probability of the document itself, regardless of class.

For classification, we compare probabilities **across different classes**.

Since $P(d)$ is the same for all classes (the document doesn't change), it acts as a constant.

Therefore, it doesn't affect which class has the maximum probability.

That's why the MAP decision rule simplifies to:

$$c_{MAP} = \arg \max P(c)P(d|c)$$